

From Screen to Presence: How Robot Embodiment Shapes Cognitive and Physiological Responses

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Abstract

Conducting Human–Robot Interaction (HRI) studies with physically present robots often involves technical, logistical, and safety constraints. As a result, researchers may prefer virtual representations of robots, such as simulations or video recordings, to create more controlled and accessible experimental settings. This study investigates whether a robot-driven cognitive task presented via video stimuli elicits responses comparable to those obtained during direct interaction with a physical robot. Twenty participants completed this task designed to induce cognitive effort under two modalities: interaction with pre-recorded robot video clips and interaction with the same robot executing identical action sequences in real time. Subjective reports, behavioral performance, and physiological signals were collected in both conditions. Behavioral, subjective, and EEG measures revealed robust effects of task difficulty that were consistent across modalities, with no significant differences between video-based and physical robot conditions. These results support the use of video stimuli as a valid, scalable, and reproducible alternative for inducing cognitive effort in HRI experiments. In contrast, other physiological modalities revealed more nuanced effects. Cardiac and electrodermal signals exhibited modality-specific modulations, independent of task difficulty, suggesting a sensitivity to the physical presence of the robot. Taken together, these findings demonstrate that, while cognitive effort studies can provide similar results with video-based and physically present robots, human perception is still impacted by the robot’s presence to a certain degree. These markers of robot presence open avenues to study presence perception in real time.

CCS Concepts

• **Human-centered computing** → **User models; User studies; Laboratory experiments; Interaction techniques**; • **Computer systems organization** → **Robotics**.

Keywords

HRI, evaluation, physiology, data acquisition, cognitive effort, physical presence

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1 Introduction

Human–Robot Interaction (HRI) research relies on controlled experimental paradigms to investigate how humans perceive, understand, and collaborate with robotic partners. However, despite the growing interest in human cognitive states during interaction, the field still lacks standardized tasks specifically designed to elicit and study targeted cognitive processes under the unique constraints of HRI. To address this gap, previous work introduced and shared three experimental paradigms inspired by neuropsychology and adapted to HRI contexts: the N-Back task [5], the Sternberg task [12], and a Cognitive Shifting task [3, 9, 16]. These tasks provided a theoretically grounded and reproducible framework for inducing controlled variations of cognitive effort through robot behavior [11].

In these studies, participants interacted with video recordings of the robot rather than with a physically present robot. This methodological choice greatly facilitated experimental reproducibility and dissemination, enabling consistent task presentation and straightforward data sharing. However, it did not account for the impact of physical robot presence on cognitive processing and user perception. Indeed, physically present robots may elicit stronger engagement than virtual agents displayed on a screen, resulting in increased attention and enhanced memory retention [6]. Such embodiment effects raise concerns about directly generalizing findings obtained from video-based paradigms to real-world HRI scenarios. The objective of this work is therefore to assess whether a previously validated cognitive effort paradigm elicits comparable subjective, behavioral, and physiological responses when performed with a physically present robot as opposed to video recordings of the same robot.



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2 Method

2.1 Task

The experiment was based on the Cognitive Shifting robotic task, designed to induce controlled variations in cognitive effort. In this task, participants were required to identify the odd one out among four sequentially presented robotic signs. Each sign corresponded to a short robot movement, and participants had to determine which sign differed from the others according to predefined parameters: localization, laterality, and movement. For instance, when the relevant parameter was laterality, if three signs were performed using the robot's left arm, the odd sign was the one executed with the right arm.

Task difficulty was manipulated across two conditions by varying the number of parameters participants needed to process. In the low-difficulty condition, identifying the odd sign required focusing on a single relevant parameter. In the high-difficulty condition, participants had to simultaneously consider two parameters, thereby increasing cognitive demand.

The task was implemented in two stimulus modalities: a video-based condition and a physical robot condition. In the video-based version, robot behaviors were presented as pre-recorded video clips displayed on a screen. Task cues and response options were also presented visually, and participants indicated their response via keyboard input. In the physical robot version, the overall task structure, cues, and response modality remained identical. The only difference was in the presentation of the robotic stimuli: instead of videos, participants observed a physically present robot executing the same action scripts used to generate the video recordings. This ensured maximum comparability between the two conditions.

2.2 Measures

The Rating Scale Mental Effort (RSME) questionnaire was employed to capture subjective perception of cognitive effort [17]. Response time (RT) and accuracy were used as behavioral measurements. As regards physiological sensors, brain activity was collected through electroencephalography (EEG) using a 20-channel Enobio (Neuroelectrics) with dry electrodes and sampled at 500 Hz. Cardiac activity sampled at 1000 Hz was collected through electrocardiography (ECG) using a Faros 360° and electrodermal activity (EDA) was collected using a Shimmer3 GSR+ Unit sampled at 128 Hz.

3 User Study

3.1 Protocol

20 participants (11 women) were recruited via advertisements disseminated on the campuses of two universities. Their mean age was equal to 27.2 (± 4.0).

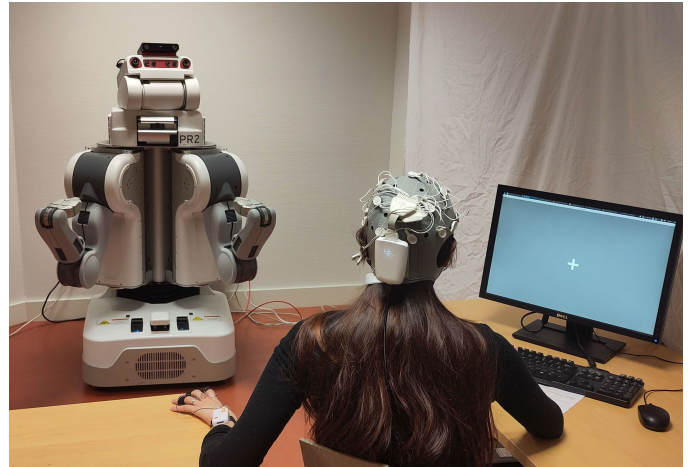


Figure 2: Setup and sensors used in this study for physiological data acquisition: an Enobio system with 20-channel, a Faros 360° device and a Shimmer3 GSR+ Unit.

Upon arrival, participants were welcomed and asked to complete an informed consent form and a demographic questionnaire. They then received a training phase consisting of three practice trials for each condition of each presentation modality. During this phase, participants were encouraged to ask questions, and the experimenter explicitly offered the possibility to repeat the training if necessary, though no participant chose to do so. After training, the physiological sensors were set up: a dry EEG cap was placed on the scalp, and EDA sensors were attached to the fingers by the experimenter. For privacy and comfort reasons, the ECG electrodes on the torso were placed by the participants themselves, following a schematic guide. The main task then began and lasted approximately one hour. It consisted of 12 blocks of 12 trials each, alternating between modalities. At the end of each block, participants completed the Rating Scale Mental Effort (RSME) questionnaire to provide a subjective measure of cognitive effort.

The block structure was organized so that participants always completed the two difficulty conditions of a given modality before moving to the next modality. The order of the two conditions within each modality was randomized. Consequently, modality switched every two blocks. To counterbalance presentation effects, half of the participants began with the video-based modality, while the other half began with the real-robot modality. At the end of the session, physiological sensors were removed, participants received monetary compensation, and the experimenter conducted a debriefing, providing a more detailed explanation of the study's goals and addressing any questions raised by the participants.

This protocol has received approval from an institutional ethics review board (project number removed for anonymity).

3.2 Analyses

Subjects were considered outliers if one of their behavioral results diverged from others of more than 3 standard deviations in at least one condition. While this rule led to the rejection during results analysis of 0 subjects, one subject was still excluded due to sensor



Figure 1: Cognitive Shifting robotic task. The elements in gray are only added in the "high difficulty" condition.

issues. Data analysis has been performed using python, with the pingouin library for statistics [14]. Given the experimental designs, a two-way RMANOVA was applied to each metric for modality effect, difficulty level effect and interactions between them. In case of a significant effect, and after testing for normality, pairwise t-tests were then performed with Bonferroni correction.

For the physiological signals, preprocessing followed a fully automated process using standard Python libraries to ensure reproducibility. EEG data were processed with MNE [4] and band-pass filtered between 1 and 100 Hz, followed by common average referencing. Resting-state EEG was further cleaned using ICA and used to train a Riemannian Artifact Subspace Reconstruction (ASR) model [2], which was then applied to task-related EEG to attenuate non-neural artifacts. EEG features consisted of spectral power in theta, alpha, and beta bands, as well as an engagement index [10]. To improve robustness, EEG features were averaged within frontal, central, and parietal regions of interest.

ECG and EDA signals were processed using NeuroKit2 [8]. Cardiac signal was filtered and peak detection was applied to extract NN intervals, from which heart rate (HR) and heart rate variability (HRV) were computed. EDA signal was filtered and decomposed into tonic and phasic components, with analyses focusing on the mean value of the peak amplitude.

All modalities were segmented using a common 14-s window spanning the four signs of each Cognitive Shifting trial.

4 Results

For subjective effort, a significant effect of task difficulty was observed ($F(1, 18) = 67.78, p < .001, \eta_p^2 = .79$). Posthoc comparisons confirmed a higher subjective effort reported in the hard condition compared to the easy condition for both stimulus modalities, with effect sizes of $\eta^2 = .40$ for the video and $\eta^2 = .34$ for the physical robot.

For the accuracy, the difficulty effect was significant ($F(1, 18) = 27.48, p < .001, \eta_p^2 = .60$). Participants performed more accurately in the easy condition than in the hard condition across both modalities, with posthoc effect sizes of $\eta^2 = .25$ for the video and $\eta^2 = .34$ for the physical robot. For the response time, the timings were significantly longer in the hard condition compared to the easy condition ($F(1, 18) = 49.26, p < .001, \eta_p^2 = .73$). Posthoc tests confirmed slower responses in the hard condition for both video and physical robot modalities, with effect sizes of $\eta^2 = .40$ in both cases.

For EEG measures, task difficulty impacted both frontal and central regions. Frontal beta power was significantly lower in the hard condition compared to the easy condition ($F(1, 18) = 10.19, p < .01, \eta_p^2 = .36$), with posthoc effect sizes of $\eta^2 = .02$ for video

and $\eta^2 = .01$ for the physical robot. Central beta also exhibited a significant difficulty effect ($F(1, 18) = 5.4, p < .05, \eta_p^2 = .23$), although only with the physical robot, with lower values in the hard condition ($\eta^2 = .01$). Similarly, the frontal engagement index decreased with task difficulty ($F(1, 18) = 9.99, p < .01, \eta_p^2 = .36$; posthoc $\eta^2 = .01$), although only the video modality reach significance.

For ECG measures, both modality and task difficulty significantly modulated heart activity. Mean heart rate showed a main effect of modality ($F(1, 18) = 9.02, p < .01, \eta_p^2 = .33$) with a higher value in the video condition. As for difficulty effect, ($F(1, 18) = 13.22, p < .01, \eta_p^2 = .42$), post-hoc comparisons confirmed higher heart rate in the hard condition compared to the easy one across both modalities (all $p < .02, \eta^2 = .01$). HRV also revealed a robust effect of difficulty ($F(1, 18) = 22.45, p < .001, \eta_p^2 = .55$), with post-hoc analyses showing a significant decrease from easy to hard conditions (all $p < .03, \eta^2 = .01-.03$).

For EDA measures, mean phasic activity showed a strong main effect of modality ($F(1, 18) = 15.79, p < .001, \eta_p^2 = .47$), with lower value in the video condition, as well as a significant interaction between modality and task difficulty ($F(1, 18) = 5.22, p < .05, \eta_p^2 = .22$). Yet, post-hoc analyses did not reach significance for this interaction.

5 Discussion & Conclusion

This study investigated whether a robot-driven cognitive task induces comparable responses when presented through video recordings or through direct interaction with a physically present robot. By addressing a central methodological question in Human-Robot Interaction (HRI), the results help clarify which behavioral and physiological responses are robust to stimulus modality and which are specifically shaped by the robot's physical presence.

At the behavioral and subjective levels, the findings were consistent across modalities. Task difficulty reliably modulated accuracy, response times, and perceived effort in both video-based and physical robot conditions, with no significant interaction between difficulty and modality. These results confirm that the Cognitive Shifting task robustly modulates classical markers of cognitive effort [7] regardless of whether the robot is physically present or displayed on a screen. Physiological measures revealed a more differentiated pattern. EEG activity, particularly frontal beta power, along with cardiac markers including heart rate (HR) and heart rate variability (HRV), consistently tracked task difficulty across both modalities, confirming their role as reliable indicators of cognitive effort in this context [1, 7]. However, HR and HRV also exhibited modality-dependent effects that were independent of task difficulty.

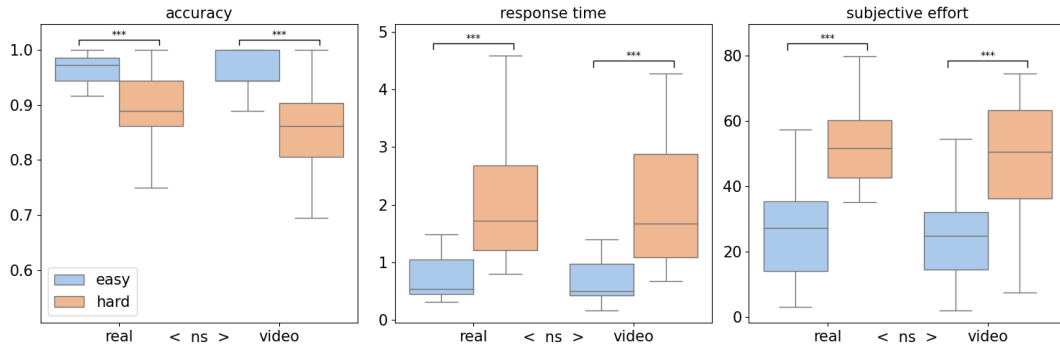


Figure 3: Results for the three non-physiological metrics. The modality effect is reported at the bottom while differences between groups are directly displayed at the boxplot level with the following rule: * ($p < 0.05$), ** ($p < 0.01$), * ($p < 0.001$), ns ($p > 0.05$).**

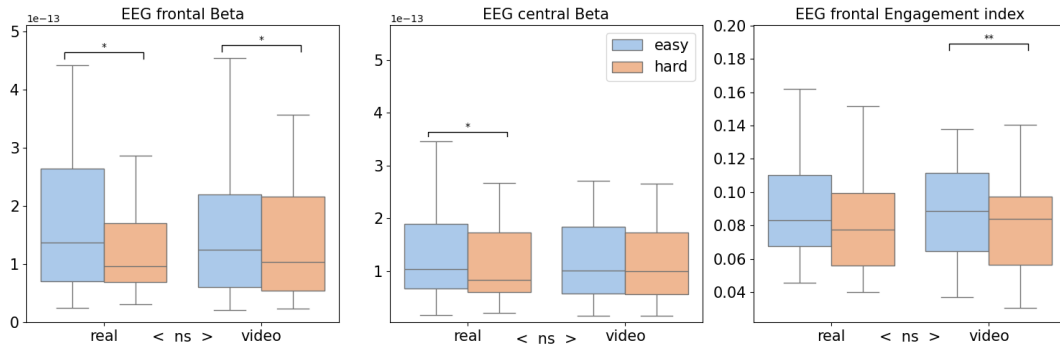


Figure 4: Results for EEG metrics. The modality effect is reported at the bottom while differences between groups are directly displayed at the boxplot level with the following rule: * ($p < 0.05$), ** ($p < 0.01$), * ($p < 0.001$), ns ($p > 0.05$).**

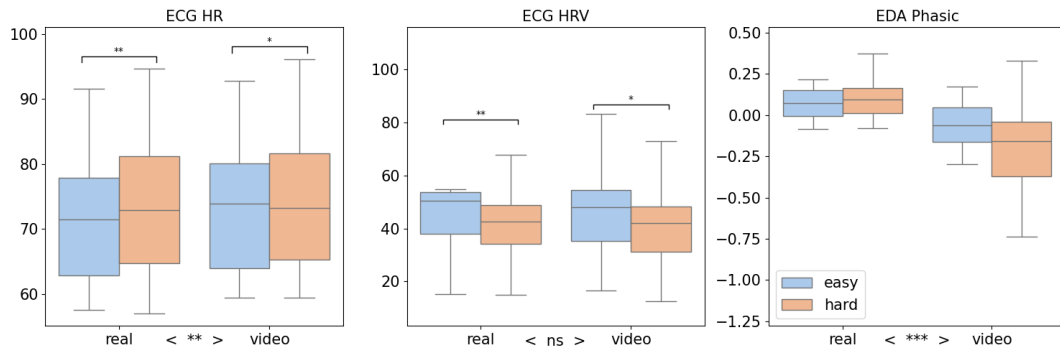


Figure 5: Results for cardiac and electrodermal metrics. The modality effect is reported at the bottom while differences between groups are directly displayed at the boxplot level with the following rule: * ($p < 0.05$), ** ($p < 0.01$), * ($p < 0.001$), ns ($p > 0.05$).**

Similar patterns observed in electrodermal activity further suggest that while cognitive effort itself is comparably engaged across modalities, the physical presence of the robot modulates additional physiological processes. These effects replicate that of studies in virtual reality [13, 15] where the same biomarkers are used to assess presence.

In conclusion, this work demonstrates that robot-driven cognitive tasks presented through videos can elicit levels of cognitive effort comparable to those induced by direct interaction with a physically present robot, as reflected by behavioral, subjective, and physiological measures. At the same time, the physical presence

of the robot produces a distinct physiological signature, observable in both cardiac and electrodermal activity, which is not tied to task difficulty. These results highlight two uses of physiological signals in HRI: as markers of cognitive effort, but also as sensitive indicators of embodiment and presence. Building on these findings, future work should investigate how presence-related physiological responses interact with key interaction variables such as physical distance to the robot, degree of anthropomorphism, and habituation over time, ultimately informing the design of more adaptive and human-centered robotic systems.

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